

# Physical reading of the exponential recession-tail fit

## Linearised Richards equation and the swale dataset

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## 1 Richards' equation in one dimension

For a homogeneous, isothermal soil column with vertical coordinate  $z$  (positive downward), volumetric water content  $\theta(z, t)$ , unsaturated hydraulic conductivity  $K(\theta)$  and matric head  $h(\theta)$ , mass conservation combined with the Buckingham–Darcy law gives the mixed-form Richards equation

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[ D(\theta) \frac{\partial \theta}{\partial z} - K(\theta) \right] - S(\theta, z, t), \quad (1)$$

where  $D(\theta) = K(\theta) \partial h / \partial \theta$  is the hydraulic diffusivity and  $S(\theta, z, t)$  is the root-uptake / direct evaporation sink. The nonlinearity is concentrated in  $D(\theta)$  and  $K(\theta)$ , both of which vary by several orders of magnitude across the unsaturated range.

## 2 Linearisation around a wetting peak

After a rainfall event the soil moisture relaxes toward an approximately uniform baseline  $\theta_b(z)$ . Write

$$\theta(z, t) = \theta_b(z) + \theta'(z, t), \quad |\theta'| \ll \theta_b. \quad (2)$$

Freezing the soil-hydraulic functions at  $\theta_b$  and dropping terms of order  $\theta'^2$  converts Eq. (??) into a linear advection–diffusion–reaction equation in  $\theta'$ :

$$\frac{\partial \theta'}{\partial t} = D_b \frac{\partial^2 \theta'}{\partial z^2} - K'_b \frac{\partial \theta'}{\partial z} - \lambda \theta', \quad (3)$$

with constant coefficients

$$D_b = D(\theta_b), \quad K'_b = \left. \frac{dK}{d\theta} \right|_{\theta_b}, \quad \lambda = \left. \frac{\partial S}{\partial \theta} \right|_{\theta_b} \geq 0. \quad (4)$$

## 3 Eigenvalue spectrum and the single-mode regime

Substituting the standard advection-removing transform  $\theta'(z, t) = \exp(\frac{K'_b z}{2D_b}) \psi(z, t)$  converts Eq. (??) into a pure diffusion equation with a constant-rate sink:

$$\frac{\partial \psi}{\partial t} = D_b \frac{\partial^2 \psi}{\partial z^2} - \left[ \frac{K_b'^2}{4D_b} + \lambda \right] \psi. \quad (5)$$

On a layer of effective depth  $L$ , the eigenvalues of the slowest decaying spatial mode (the value of which depends on the boundary conditions chosen at  $z = 0$  and  $z = L$ ) take the canonical form

$$\boxed{\frac{1}{\tau_n} = \underbrace{D_b \left( \frac{n\pi}{L} \right)^2}_{\text{vertical diffusion mode } n} + \underbrace{\frac{K_b'^2}{4D_b}}_{\text{gravity advection}} + \underbrace{\lambda}_{\text{ET sink}}}, \quad (6)$$

where  $n = 1, 2, 3, \dots$ . The recession is dominated by the fundamental mode ( $n = 1$ ) within a few  $\tau_1$  of the peak. With mixed Dirichlet–Neumann boundaries (closed surface, fixed- $\theta$  bottom into a water table) the prefactor on the diffusion term reads  $\pi^2 D_b / (4L^2)$  instead of  $\pi^2 D_b / L^2$ . Either way, the *form* of Eq. (??) is the same: a single characteristic timescale built additively from three physical processes.

**Diagnostic.** If a recession is well described by a single  $\tau$  (high  $R^2$  for  $\theta'(t) = A e^{-t/\tau} + C$ ) and the corresponding power-law exponent  $\alpha$  is near zero, the soil is in this single-eigenmode regime. Higher modes have already decayed and no heterogeneity-induced heavy tail is visible.

## 4 Comparison with alternative tail models

Different physical regimes produce qualitatively different recession shapes:

| Regime                                        | Recession shape               | Diagnostic exponent              |
|-----------------------------------------------|-------------------------------|----------------------------------|
| Linearised Richards (single mode)             | $\theta \propto e^{-t/\tau}$  | $\alpha \approx 0$               |
| Capillary diffusion, semi-infinite half-space | $\theta \propto t^{-1/2}$     | $\alpha \approx 0.5$             |
| Boussinesq groundwater, late time             | $\theta \propto t^{-2}$       | $\alpha \approx 2$               |
| Brutsaert–Nieber, $dQ/dt = -aQ^b$             | $\theta \propto t^{-1/(b-1)}$ | $\alpha = 1/(b-1)$               |
| Distribution of pore-drainage times           | $\sum_i A_i e^{-t/\tau_i}$    | apparent power-law over a window |

For the swale data the median fits give  $\alpha = 0.006$ – $0.04$  and exponential  $R^2$  consistently above the power-law  $R^2$  across all (depth, treatment) cells. This places the system in the single-mode linear-reservoir regime and justifies using  $\tau$  as the primary descriptor.

## 5 Inverting the fitted $\tau$ for soil parameters

When the three contributions to Eq. (??) cannot be separated from a single trace, the fitted  $\tau$  provides only an *upper bound* on the diffusion timescale:

$$D_b \leq \frac{L^2}{\pi^2 \tau}. \quad (7)$$

For  $L = 10$  cm and the swale-dataset medians:

| Depth | Treatment | $\tau$ (h) | $\tau$ (s)        | $D_b$ upper bound ( $\text{m}^2 \text{s}^{-1}$ ) |
|-------|-----------|------------|-------------------|--------------------------------------------------|
| 10 cm | control   | 56         | $2.0 \times 10^5$ | $5.0 \times 10^{-9}$                             |
| 10 cm | swale     | 85         | $3.1 \times 10^5$ | $3.3 \times 10^{-9}$                             |
| 40 cm | control   | 73         | $2.6 \times 10^5$ | $3.9 \times 10^{-9}$                             |
| 40 cm | swale     | 3261       | $1.2 \times 10^7$ | $8.6 \times 10^{-11}$                            |

The control 10 / 40 cm numbers ( $\sim 10^{-9} \text{ m}^2/\text{s}$ ) sit in the canonical range for unsaturated loam near field capacity. The swale 10 cm bound is  $\sim 30\%$  lower than control. The swale 40 cm bound is two orders of magnitude smaller than control 40 cm and cannot be reconciled with any realistic  $D_b$ : it is the signature of a regime where the diffusion mode is not the rate-limiter at all.

## 6 Why the swale 40 cm $\tau$ blows up

Eq. (??) says  $1/\tau$  is a sum of non-negative contributions, so  $\tau \rightarrow \infty$  only when *all three* collapse simultaneously:

- the diffusion mode is suppressed (a saturated boundary clamps  $\theta'$  from below, or a low-permeability layer blocks vertical flux);
- gravity advection  $K'_b$  vanishes (the gravitational gradient is balanced by a continuous lateral inflow that re-supplies water faster than gravity drains it); and
- the ET sink  $\lambda$  is small (deep root demand is weak at 40 cm, or the available water in the perched layer is large compared to weekly demand).

Of these three, sustained lateral inflow from upslope mounds is the most parsimonious single explanation, and is the one we can test independently: it predicts a wetting-front lag at 40 cm that *matches* 10 cm of the same plot rather than the 40 cm of the control plot.

## 7 What a Richards-model inversion would add

The linear-reservoir model collapses three independent physical processes into a single observable  $\tau$ . To separate them one needs either:

1. A *vertical-profile constraint*: two depths in the same column with shared  $D_b$  but different surface coupling. Each depth provides one equation in  $\{D_b, K'_b, \lambda\}$ . The current 10 cm & 40 cm pair already gives this.
2. An *independent ET estimate*: ATMOS 14 vapour-pressure deficit and air temperature constrain  $\lambda$  separately. We have these data on logger Z6-19570.
3. A *forward Richards solver* (e.g. HYDRUS-1D / phydus) with the measured rainfall as upper boundary and observed  $\theta(z, t)$  as the inversion target, returning van Genuchten / Mualem retention parameters  $(\theta_r, \theta_s, \alpha_{vG}, n, K_s)$  per layer and treatment. Option (3) subsumes (1)–(2) and quantifies the lateral-inflow hypothesis as a residual.